TEXORA · LECTURE SHEET

Torsion

Twisting shafts: how torque becomes shear stress proportional to radius, the torsion formula and the angle of twist, power from torque and speed, solid versus hollow sections — and the sizing calculation every line shaft, spindle and drive in the mill depends on.

ME-3210 · SOLID MECHANICS AND MACHINE DESIGN · TOPIC 3 · LEVEL 3 TERM 2

1. The torsion formula

Apply torque T to a circular shaft fixed at one end. Cross-sections rotate rigidly about the axis; a surface line twists into a helix. Shear strain at radius r grows in proportion to r (zero at the axis, maximum at the surface), and since τ = Gγ inside the elastic limit, shear stress inherits the same distribution:

$$\tau = T\,r / J, \quad \text{worst at the surface: } \tau_{max} = T\,R / J, \quad J = (\pi/32)(D^4 - d^4)$$

J is the **polar second moment of area** (for a solid shaft d = 0, $J = \pi D^4/32$). Two consequences matter daily: stress depends on the *fourth power* of diameter, so small size changes transform capacity; and material near the axis carries almost no stress — which is why hollow shafts exist.

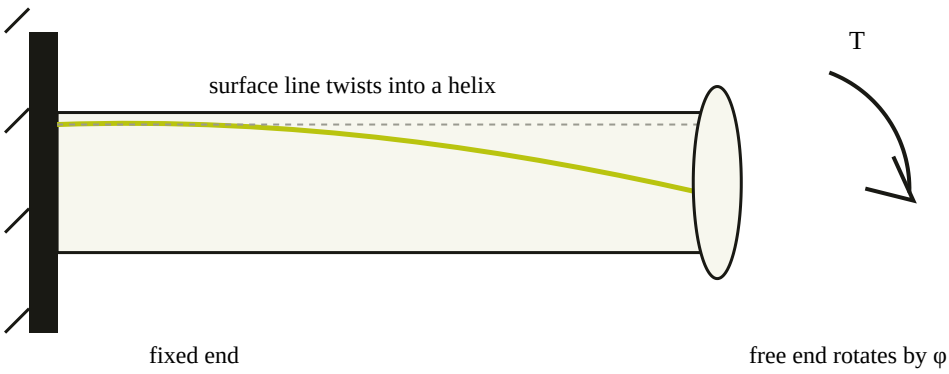


Figure 1 — A shaft under torque: the fixed end cannot rotate; the free end twists through angle φ, and shear stress grows linearly from axis to surface.

2. Angle of twist and power

The rotation accumulated along the shaft is as design-critical as the stress — a long shaft can be strong enough yet twist too much for the machine it drives (timing, register, cam phasing):

$$\varphi = TL/(GJ) \quad \text{(radians)}$$

Torque and power trade through speed: $P = T\omega = 2\pi NT/60$, with N in rpm — the conversion between a motor nameplate (kW at rpm) and the shaft's working torque.

WORKED EXAMPLE — STRESS AND TWIST OF A SOLID SHAFT

A solid shaft of 50 mm diameter transmits 20 kW at 300 rpm (G = 80 GPa). Torque: $\omega = 2\pi \times 300/60 = 31.42$ rad/s, $T = 20{,}000/31.42 =$ **636.6 N·m**. $J = (\pi/32)(0.05)^4 = 6.136 \times 10^{-7}$ m⁴. Surface stress: $\tau = TR/J = 636.6 \times 0.025/6.136 \times 10^{-7} =$ **25.9 MPa** — comfortable against a 45 MPa allowance. Twist over 2 m: $\varphi = TL/(GJ) = 636.6 \times 2/(80 \times 10^9 \times 6.136 \times 10^{-7}) = 0.0259$ rad = **1.49°** — strong enough, and stiff enough for most mill drives; a printing or knitting drive with register requirements might still reject it.

WORKED EXAMPLE — SIZING A SHAFT FROM ALLOWABLE STRESS

Design a solid shaft for 50 kW at 250 rpm with $\tau_{allow} = 45$ MPa. $T = 50{,}000/(2\pi \times 250/60) = 50{,}000/26.18 =$ **1909.9 N·m**. From $\tau = 16T/(\pi d^3)$: $d^3 = 16T/(\pi\tau_{allow}) = 16 \times 1909.9/(\pi \times 45 \times 10^6) = 2.162 \times 10^{-4}$ m³ → d = **60.0 mm**. Specify 60 mm — and then check the twist and any keyway weakening (Topic 5) before signing it off.

3. Solid versus hollow, and shafts in combination

A hollow shaft with outside diameter D and inside 0.6D has $J = (\pi/32)D^4(1 - 0.6^4) = 0.87$ of the solid's J but only $(1 - 0.36) = 0.64$ of its weight — nearly the same strength for a third less material, which is why propeller shafts, axles and long mill line shafts are bored. **Series** shafts (end to end) carry the same torque and add their twists; **parallel** (statically indeterminate) shafts — a shaft fixed at both ends with torque applied between — split torque in proportion to GJ/L of each segment. Stepped shafts concentrate stress at shoulders: fillets are not decoration (Topic 5 again).

4. Key points

- $\tau = Tr/J$ with $J = (\pi/32)(D^4 - d^4)$: stress ∝ radius, capacity ∝ diameter⁴.
- $\varphi = TL/(GJ)$: stiffness is a separate check from strength — the worked shaft passes strength at 25.9 MPa yet twists 1.49° over 2 m.
- $P = 2\pi NT/60$ converts nameplate power and speed into working torque.

- Sizing: $d^3 = 16T/(\pi\tau_allow)$ for solid shafts — 50 kW at 250 rpm and 45 MPa lands exactly on 60 mm.
- Hollow ($d = 0.6D$) keeps 87% of the strength for 64% of the weight.

5. Practice

Answer in your own words; brief model answers follow.

1. A solid shaft 40 mm in diameter carries 500 N·m. Find the maximum shear stress.
2. What torque can a 60 mm solid shaft carry if $\tau_allow = 50$ MPa? What power at 400 rpm?
3. A hollow shaft has outside 80 mm, inside 50 mm, and carries 3 kN·m. Find the outer-surface shear stress.
4. The shaft of Question 1 is 1.5 m long ($G = 80$ GPa). Find its angle of twist in degrees.
5. A shaft fixed at both ends (each segment 1 m, same diameter and material) has a torque of 900 N·m applied at the junction. Find the reaction torques.
6. Why does doubling a shaft's diameter multiply its torque capacity eightfold but its stiffness sixteenfold?

Answer key

1. $\tau = 16T/(\pi d^3) = 16 \times 500/(\pi \times 0.04^3) = 8000/2.011 \times 10^{-4} = 39.8$ MPa.
2. $T = \tau\pi d^3/16 = 50 \times 10^6 \times \pi \times (0.06)^3/16 = 2120$ N·m. $P = 2\pi \times 400/60 \times 2120 = 88.8$ kW.
3. $J = (\pi/32)(0.08^4 - 0.05^4) = (\pi/32)(4.096 \times 10^{-5} - 6.25 \times 10^{-6}) = 3.408 \times 10^{-6}$ m⁴. $\tau = TR/J = 3000 \times 0.04/3.408 \times 10^{-6} = 35.2$ MPa.
4. $J = (\pi/32)(0.04)^4 = 2.513 \times 10^{-7}$ m⁴. $\varphi = TL/(GJ) = 500 \times 1.5/(80 \times 10^9 \times 2.513 \times 10^{-7}) = 0.0373$ rad $= 2.14°$.
5. Equal twist compatibility: the junction rotates so that $T_1L/(GJ) = T_2L/(GJ)$ with equal L, G, J → $T_1 = T_2 = 450$ N·m each.
6. Capacity: $\tau = 16T/(\pi d^3) \rightarrow T \propto d^3$ at fixed τ — doubling d multiplies capacity by $2^3 = 8$. Stiffness: $\varphi \propto 1/J \propto 1/d^4$ — doubling d divides twist by $2^4 = 16$.